

Program : Diploma in Food Processing Technology	
Course Code : 2421	Course Title: Fundamentals of Food Processing Technology
Semester : 2	Credits: 3
Course Category: Engineering science	
Periods per week: 3 (L: 3 T: 0 P: 0)	Periods per semester: 45

Course Objectives:

- To impart knowledge on different macro and micro nutrients present in food
- To learn the importance of macro and micro nutrients present in food
- To impart knowledge on classification of macro and micro nutrients

Course Prerequisites:

Topic	Program / Course Name
Basic knowledge of chemistry	Secondary school

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the form and properties of water in food	10	Understanding
CO2	Illustrate the composition and classification of carbohydrates	13	Understanding
CO3	Describe the structure and categories of protein in food	12	Understanding
CO4	Explain the importance of micro nutrients and lipids	10	Understanding

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	CognitiveLevel
CO1	Explain the form and properties of water in food		
M1.01	List the different macro and micro nutrient present in food	4	Understanding
M1.02	Describe the Water activity and its importance in food	3	Understanding
M1.03	Illustrate the different form, structure and properties of water in food	3	Understanding
Contents: Food -definition. Macro nutrient-definition, types. Micro nutrients-definition and type. Water activity-definition, importance, measurement. Water structure-forms in food Three forms (Free, adsorbed, bound), properties.			
CO2	Illustrate the composition and classification of carbohydrates		
M2.01	Describe the chemistry of carbohydrates	3	Understanding
M2.02	Outline the composition and structure of carbohydrates	3	Understanding
M2.03	Explain the classification and properties of carbohydrates	3	Understanding
M2.04	Illustrate various sources and functional importance of carbohydrates	3	Understanding
	Series Test – I	1	

Contents : Carbohydrates-definition,chemistry,importance-composition,structure of glucose, fructose, galactose- classification (based on structure)-Properties-various sources-functional role			
CO3	Describe the structure and categories of protein in food		
M3.01	Describe the functional role of protein	4	Understanding
M3.02	Explain the structure of protein	4	Understanding
M3.03	Illustrate the type of classification of protein	4	Understanding
Contents: Protein- definition- importance-functional role-structure (primary, secondary, tertiary, quaternary) -classification based on composition (simple, conjugated, derived), structure (fibrous, globular, intermediate) Function (structural protein, enzymes protein, transport protein, storage protein, genetic protein)			
CO4	Explain the importance of micro nutrients and lipids		
M4.01	Explain the classification and properties of lipids	3	Understanding
M4.02	Illustrate the types and sources of vitamins	3	Understanding
M4.03	Describe the types and functional role of minerals	3	Understanding
	Series Test – II	1	
contents: Lipids definition, classification, properties. Vitamin definition, classification, sources. Minerals- types, role, importance in food.			

Text / Reference

T/R	Book Title / Author
R1	Sivasankar, B, “Food processing and preservation” Prentice – Hall of India Pvt.Ltd.New Delhi 2002.
R2	Srinivasan Damodaran, Kirk L. Parkin, and O.R. Fennema, E, “Food Chemistry”4thEdition, CRC Press, New York 2007
R3	Chopra, H.K. and P.S. Panesar. “Food Chemistry”. Narosa, 2010

R4	Food Chemistry, H.-D. Belitz, W. Grosch, P. Schieberle 3rd ed. 2004, Publisher:Springer
R5	Charley, H, “Food Science” John Wiley and Sons Inc., New York, 1982.
R6	Birch, G.G., Brennan, J. G. and Parker, K. J, “The Sensory Properties of Foods”Applied Science Publication, London, 1977.
R7	Robinson, D. S, “Food – Biochemistry and Nutritional Value” Longman Scientificand Technical, London, 1987.
R8	Handbook of Food Chemistry, Peter Chi Keung Cheung, Bhavbhuti M. Mehta, 1sted. 2015, Springer Berlin Heidelberg

Online Resources

Sl.No	Website Link
1	https://health.clevelandclinic.org/macronutrients-vs-micronutrients/
2	https://www.getsmarter.com/blog/market-trends/what-are-macronutrients-and-micronutrients/
3	https://www.healthline.com/health/food-nutrition/micros-vs-macros
4	n.video.search.yahoo.com/yhs/search?fr=yhs-omr-001&ei=UTF-8&hsimp=yhs-001&hspart=omr¶m1=y6bdVFVIsvuYsgECIQfz8NEPSp4FWG51g5cOG5gIsG 4PGGuwc
5	https://egyankosh.ac.in/bitstream/123456789/12249/1/Unit-4.pdf